

Solving SAT in polynomial time?

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Abstract

Our goal is to shed some light on the P vs NP problem by looking at a SAT solver and investigate whether it runs in polynomial time. Given the NP-completeness of SAT, if it does, this would imply $P = NP$.

Keywords: P vs NP, polynomial time SAT solver, Boolean algebra We would like to acknowledge all of the many people who contributed to this work.

1 A SAT solver which might run in polynomial time

1.1 Input Boolean w

Either $' = p_i$ for a propositional variable p_i or $' = : ' _ 1$ for some Boolean w $' _ 1$, or $' = ' _ 1 \wedge ' _ 2$ for Boolean w s $' _ 1$ and $' _ 2$, or $' = ' _ 1 _ ' _ 2$ for Boolean w s $' _ 1$ and $' _ 2$.

1.2 Compute a polynomial associated with ' called (') as follows:

$$(p_i) := x_i$$

$$(: ' _ 1) := 1 + (' _ 1)$$

$$(' _ 1 \wedge ' _ 2) := (' _ 1) (' _ 2)$$

$$(' _ 1 _ ' _ 2) := 1 + (1 + (' _ 1))(1 + (' _ 2))$$

Calculations are done in F_2 .

1.3 Let $J = \langle h + 1; x_1^2 + x_1; \dots; x_n^2 + x_n \rangle$ be an ideal in $F_2[x_1; \dots; x_n]$

Note that $J = \langle h + 1; x_1^2 - x_1; \dots; x_n^2 - x_n \rangle$.

- 1.4 Find a Gröbner basis G for J which might be done in polynomial time using the FGLM algorithm or a different algorithm
- 1.5 Check whether $\bar{1}^G = 0$
 - 1.5.1 If $\bar{1}^G = 0$, then φ is not satisfiable
 - 1.5.2 If $\bar{1}^G \neq 0$, then φ is satisfiable